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Concept for Introducing Sustainable Product Service Systems through a Business Game

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Abstract

A fundamental change in how businesses are conducted and operated is an essential factor to reach more sustainable solutions on the industry level. However, the education of graduate students in Product Service Systems (PSS) or specifically in sustainable PSS is still very limited. Despite the fact that there has been significant work on how PSS performs as a business concept and how it may contribute to competitiveness, the sustainability-orientated PSS research has paid limited attention to the education sector. Therefore, this paper presents a concept on how sustainable PSS can be taught to production management students using a business game. The presented concept is based on a business game currently used in production technology education at the University of Applied Sciences Emden/Leer.

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1. Introduction

The Brundtland Commission (1987) defined sustainability as "development that meets the needs of the present without compromising the ability of future generations to meet their own needs" [1]. With the increase in human population, the demand for products rises. Mass production is the current way to meet this rising demand, but this form of production consumes a great amount of resources, like energy, non-renewable materials, renewable materials, ancillary inputs, and fossil fuels. Additionally, significant amounts of waste, effluents, and emissions are generated during each stage of the product life cycle [2]. By transforming businesses to include environmental as well as social targets while maintaining economic competitiveness, business model innovation is proposed as key to more sustainable development [3]. A fundamental change in how businesses are conducted and operated is an essential factor to reach more sustainable solutions on the industry level. Education plays a major role in developing more sustainable solutions in companies. Product Service Systems (PSS) are one opportunity to increase the sustainability performance of enterprises.

Most literature for PSS in education from previous Conferences on Industrial Product-Service Systems refers to the qualifications of employees in companies. Nguyen, Schürmacher, and Stark (2014) present an approach for introducing the methods and tools of the PSS development methodology in companies. The approach contains methods and tools that sensitize corporate management and train project managers and employees [4]. Nemoto et al. (2014) developed an educational business game to familiarize users with the philosophy of PSS and changing their mindset. The business game is specifically directed towards companies' designers [5]. Süße, and Wilkens (2014) propose a game-based learning scenario in order to integrate users in PSS work environments. The concept provides a context-rich learning environment while focusing on key challenges of the process of transformation of PSS [6]. Peillon, Medini, and Dubruc (2016) argue that if in-company training is essential to

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the diffusion of knowledge and practices of PSS, one should not forget that initial education is a key point, as it reduces the number of subsequent in-company training programs needed. In this regard, they present an original training program that was built and implemented at the Ecole des Mines de Saint-Etienne (EMSE). The main characteristics of this program are multidisciplinary, in terms of mixing PSS engineering and managerial approaches and mixing theory, case studies, and the use of simulation, based on several educational tools [7]. However, the education of graduate students in PSS, or specifically in sustainable PSS, is still very limited. Despite the fact that there has been significant research on how PSS performs as a business concept and how it may contribute to competitiveness [3], the sustainability-orientated PSS research has paid limited attention to the education sector.

Therefore, this paper presents a concept of how sustainable PSS can be taught to production management students using a business game.

Business games were developed in the military and business sectors to qualify leaders in multifaceted and practice-oriented learning environments. This learning tool became a common method in schools and universities to prepare students working in "real life" situations. The advantage of learning in a simulated environment (business game) is that the effects of decisions and actions can be experienced in a safe, fast-moving, and inexpensive environment; mistakes are allowed, and a restart of the game is always possible [8]. For this approach, a business game currently used in production and logistics education at the University of Applied Sciences Emden/Leer is presented and how it can be enhanced to meet the challenges of sustainable PSS and current trends for digitalization. The digitalization part is introduced through the project "Automated Classroom" which aim to create a coherent industry 4.0 learning environment.

After this Introduction, Section 2 describes the theoretical background for PSS and sustainable PSS, the general procedure for business gaming and provides a description of the "Automated Classroom" project. Section 3 introduces the concept of the currently used business game known as the "Christmas Production Game." Based on this concept, Section 4 presents options for changing the current business game to address the concepts of sustainable PSS. Section 5 contains the conclusion and the outlook of the paper followed by the references.

2. Theoretical Background

This section presents background information for performing business games, widely accepted definitions for PSS, and a description of the "Automated Classroom" project.

2.1. Procedure for successful simulation gaming

Business games, also called simulation games or serious games, allow for active training of students in a realistic but safe environment that expands their abilities with new knowledge [9]. Business games as game-based learning for education offer advantages such as a holistic understanding of the system, new perspectives on problems in a system, and an understanding of relationships among actors in a system [10]. The process of receiving information and applying knowledge by doing is almost immediate. The retention rate of knowledge is high in comparison to traditional teaching approaches, such as lecturing, demonstrating, and group work. There exists no strict procedure for business games. Von Anmeln, and Kramer (2016) divided simulation gaming into four phases [10]:

1. Planning
2. Preparing
3. Action
4. Evaluation

In the planning phase, the gamemaster selects a suitable business game. The following aspects must be considered in the selection process: the number of participants playing the business game, the learning targets, the participants' background knowledge and skills, the participants' experience with simulation gaming, and the languages required to play the game. In the preparing phase, the gamemaster introduces the participants to the targets, the rules, available and/or allowed tools, and information about the business game. Based on this introduction, the participants prepare the needed tools and information for the business game in order to achieve the targets. In the action phase, the participants play the business game. The gamemaster controls the game and, gives new information (e.g., new production orders) to participants and initiates problems (e.g., breakdown of tools) for the participants at specific time intervals. In the evaluation phase, the participants discuss achievements and problems experienced during the action phase. Based on this discussion, the participants develop improvements to achieve the targets, which can be used to play the game a second time.

2.2. Definition and types of PSS and sustainable PSS

There exists no standard definition for PSS or sustainable PSS. Goedkoop, Halen, Riele, and Rommens define PSS as "a marketable set of products and services capable of jointly fulfilling a user's need. The PSS is provided by either a single company or by an alliance of companies. It can enclose products (or just one) plus additional services. It can enclose a service plus an additional product. And product and service can be equally important for the function fulfilment" [11, p. 18]. Moreover, Vezzolo (2018) defined sustainable PSS as "models providing an integrated mix of products and services that are together able to fulfil a particular customer demand (to deliver a unit of satisfaction), based on innovative interactions between the stakeholders of the value production system (satisfaction system), where the economic and

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competitive interest of the providers continuously seeks environmentally beneficial new solutions" [12].

However, various classifications have been proposed for PSS, which have varied effects on the sustainability performance of companies. According to Tukker (2004), PSS can be classified in three main categories [13]:

- product-oriented services,
- use-oriented services, and
- result-oriented services.

The first main category is product-oriented services. Here, the business model is still mainly geared towards the sale of products, but some extra services are added [13]. Regarding environmental sustainability in product-oriented services, only effects through more quality maintenance could be an outcome, but selling as many products as possible remains the business incentive [3]. The second main category is use-oriented services. Here, the traditional product still plays a central role, but the business model is not geared towards selling products. The product stays in ownership with the provider and is made available in a different form, sometimes shared by several users [13]. In use-oriented services, such as leasing, there may be a negative effect in that products are treated with less care by the customer, while there is a chance to improve resource usage, such as in car-sharing systems. Social effects include increased availability of a product for individuals who cannot afford to own a product, e.g., personal mobility for people who cannot afford to own a car but have access to a car through a sharing provider [3]. The last main category is result-oriented services. Here, the client and provider in principle agree on a result, and there is no pre-determined product involved [13]. Result-oriented services have until today been the most unconventional PSS and have a large potential for sustainability benefits, since material resources are only a cost factor for the provider and greater production does not lead to increased revenue. However, each case has to be evaluated individually, since simply outsourcing, e.g., providing lunch to a catering service. might only have a marginal efficiency effect [13].

2.3. Project: Automated Classroom

The "Automated Classroom" project is an internally financed project at the University of Applied Sciences Emden/Leer and was initiated in 2017 [14]. The aim of the project is to create a coherent industry 4.0 learning environment, which makes the challenges and opportunities of digitization in the interaction of the individual assets in industry tangible and perceptible for students. Students, scientific employees, and professors from different programs of study (e.g., Master of Mechanical Engineering, Bachelor of Industrial Business Systems, Master of Industrial Informatics, Business Administration, etc.) carry out the project. The "Automated Classroom" consists of different physical processes for the production and transport of products (laser engraver, 3D printer, transport robots) and digital systems for production management, e.g., an enterprise resource management (ERP) system. The main project tasks include the implementation of these physical processes and digital systems; the development and configuration of generic industry 4.0-compatible interfaces, which connect the physical systems with digital systems; and the possibility of implementing different production scenarios to present new market offers (products and services and subsequent business models) resulting from the various new Industry 4.0 technical possibilities. The ERP system is a core component of the "Automated Classroom," as it manages the production. Customer and production orders are recorded and managed via this system. Moreover, it serves the other systems as a source of information about open orders and production steps to be processed. In the case of the automated classroom, machines or users can retrieve this information via web services from the ERP underlying database. The interface is based on a software adapter that implements all relevant web services for the query and feedback of products and production data and translates these into corresponding database queries for the Transfact ERP system.

3. Concept of the Current Business Game

At the University of Applied Sciences Emden/Leer, business gaming is a method that has been applied in teaching for years. Some of the games are developed at the University (e.g., SKAI-Lab), while others are purchased from game developers such as Topsim™ Tata and PRESTO. The games are part of the business and technical program curricula. For the lecture Introduction to ERP/PPC Systems, the business game "Christmas Production" was developed by a research group from the University of Applied Sciences. The main goal of the business game is for students to learn how to set up and use an ERP system to manage a production system, from order acceptance to shipping the products to customers (order-to-ship cycle). To pass the class, the students must complete the business game, including ERP system set up, planning and operating the production, and evaluating their actions and results. The following subsections describe the framework conditions and the game environment, the task for the students, and an example of how the game can be played.

3.1. Framework conditions and game environment of the "Christmas Production Game"

Requirements for the business game include knowledge about using the ERP system Transfact (www.transfact.de). Usually, students acquire the required knowledge while taking a course with lectures and hands-on training on the Transfact system. Examples of such courses are Introduction to ERP/PPC Systems and Sustainable Production.

The background story for the business game is as follows: A newly founded startup wants to introduce an ERP system for their production of Christmas-related products. To test the

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ERP system, the business cycle from customer order to shipment of the product (order-to-ship cycle) is to be handled. In 2018, the startup decided to produce and sell chains of light for Christmas decoration (see Figure 1 for an example). The sale of the Christmas light chains follows a conventional business model: sell as much as possible without extra services for the customers. The Christmas light chains consist of wooden houses and can be produced and sold with various colored roofs (natural, red, gold, and silver) and various colored inlays (without inlay, red, and orange). The Christmas light chains come packaged in a red gift bag. All options can be delivered with or without a greeting card and as an assembly kit or as a finished product. There are four customers for the products: a test consumer, Santa Claus GmbH, Angel-Shipping GmbH, and Gnome AG. Additionally, the products can be cash-sold directly to end customers at the factory outlet. The startup management forecasts 80 finished Christmas light chains in different variations and 20 assembly kits per day. To produce and sell the products, the startup has four permanent employees and the option to employ temporary workers (students from the nearby university).

Fig. 1. Example of one possible product variation.

3.2. Tasks for the students

Based on the procedure for business games described in Section 2.1 and the framework conditions and game environment described in Section 3.1, the students have three main tasks.

In task 1, the students must prepare the ERP system to reflect the Christmas light chain business from order acceptance to delivery of the light chains (order-to-ship cycle). In task 2, the business game is physically carried out, planned, and controlled with the aid of the ERP system. According to the game's structure, each assembly station must be prepared with the required resources and tools by the students. The lead time (order acceptance to delivery) of the products is limited according to the set delivery time. This is calculated and controlled by the gamemaster with the aid of the ERP system. In task 3, the students evaluate their results from tasks 1 and 2 to create and discuss improvements to produce the light chains more efficiently.

3.3. Example game solution

Figure 2 shows an example production structure for the business game. The production consists of the following assembly stations: assembly station, paint shop, quality control/rework, package station, and shipping. In addition to these assembly stations, two employees work in customer service and as production schedulers. Based on this structure, task 2 of the business game starts with a 30-min preparation phase. In this phase, the students prepare all assembly stations with the required resources and tools. Based on the forecast of expected sales, the production-planning team gives production orders to the assembly station to produce the Christmas light chains. These light chains are temporarily stored at the assembly station and documented in the ERP system. After 30 minutes, the customer service station receives customer orders via telephone or email from the customers (orchestrated by the business gamemaster). Each customer order contains information about the costumer name, the number and type of light chains needed, and the delivery time.

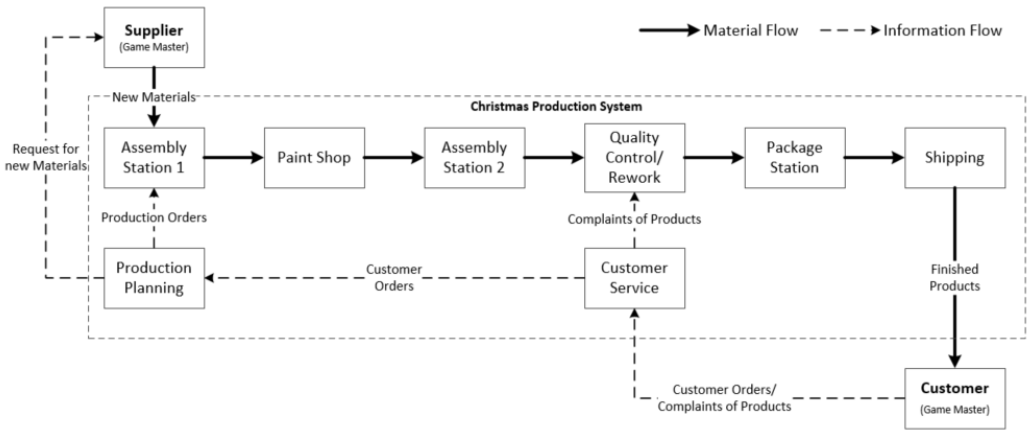

Fig. 2. Example of one possible game solution of the current business game.

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The customer service employee records the customer orders in the ERP system, which monitors the inventory of assembly kits, finished products, colors, gift bags, and greeting cards in storage. In case of material shortages, the system sends alerts to the production-planning team to initiate production or purchase orders. If enough material is in storage, the production orders are released to the shop floor. The production order is presented to the assembly station, which assembles houses and passes them on to the paint shop. Based on the production order, the paint shop colors the houses, passes the colored and dried items on to the second assembly station to join the houses with the chain of light, and then forwards them to the quality control/rework station to check the quality of the assembly, quality of the paint, and completeness of the order. If the light chains pass the quality check, the package station packs the light chains based on the production order in gift bags and, if needed, additionally with a greeting card. In the last step, the shipping station generates invoices and delivery notes in the ERP system and sends off the packed light chains with the documents to the customer. If the quality of the light chains is not acceptable and/or the order is incomplete, the customer sends the light chains back to customer service to handle the issues.

4. Development Toward a Business Game Teaching PSS and Increasing Sustainability

The following section presents an extended version of the simulation along with an example of how sustainable PSS can be taught using a business game. The section follows the same structure as Section 3 and presents the main differences and challenges of the original "Christmas Production Game" in comparison to the "PSS Christmas Production Game."

4.1. Framework conditions and game environment of the "PSS Christmas Production Game"

In addition to the requirements for playing the original business game, the students need basic knowledge about sustainability and PSS to understand the benefits for companies.

The background story is changed as follows for the business game: a newly founded startup decides to use a specific ERP system to map the Christmas present business from order acceptance to delivery of the presents (order-to-ship cycle). To differentiate from competitors, the startup attaches importance of sustainable products with individual features. In 2018, the startup decided to produce and sell Christmas light chains. The Christmas light chains can be customized through personal engravings and 3D-printed greeting cards (see example in Figure 3). To save labor and development time and costs, an external provider is assigned to create the personal engravings and 3D-printed greeting cards. The external provider is played by the gamemaster, who operates the "Automated Classroom." The "Automated Classroom" simulates a result-oriented PSS in the business game.

Fig. 3. Example of customized products offered by the "Automated Classroom."

In addition to the conventional business model, the startup decides to offer rentals as a new business model. Customers can rent the light chain for special events. This offer includes the opportunity for limited personal engravings. This additional business model simulates a use-oriented PSS in the business game.

4.2. Updated tasks for the students

The tasks for the students in the extended version of the business game are changed as follows: In task 1, the students must prepare the ERP system and production system to reflect the Christmas light chain business from order acceptance to delivery of the light chains (order-to-ship cycle) as PSS. In task 2, the business game is physically carried out, planned, and controlled with the aid of the ERP system. The requirements for lead time and assembly stations are the same as in the original version. In addition to the required actions for improvements in task 3, recommendations from the students on how the startup can increase its sustainable performance are expected.

4.3. Example game solution

Figure 4 presents an example production structure for the extended version of the business game. The new parts are marked in orange (result-oriented PSS) and green (use-oriented PSS).

For the result-oriented PSS (orange mark), the Christmas light chain production is changed as follow: In case customers order a personal engraving and/or a 3D-printed greeting, the production scheduler orders the features via a web shop from the "Automated Classroom." Based on the customers' orders, the production scheduler must upload or choose a picture for the engraving and/or a text for the greeting card. The "Automated Classroom" automatically processes the order. The gamemaster assistant is responsible for the control of the processes and the commissions with new materials. Via the robot transport system, the finished customized products are provided by the assembly station for further production processes.

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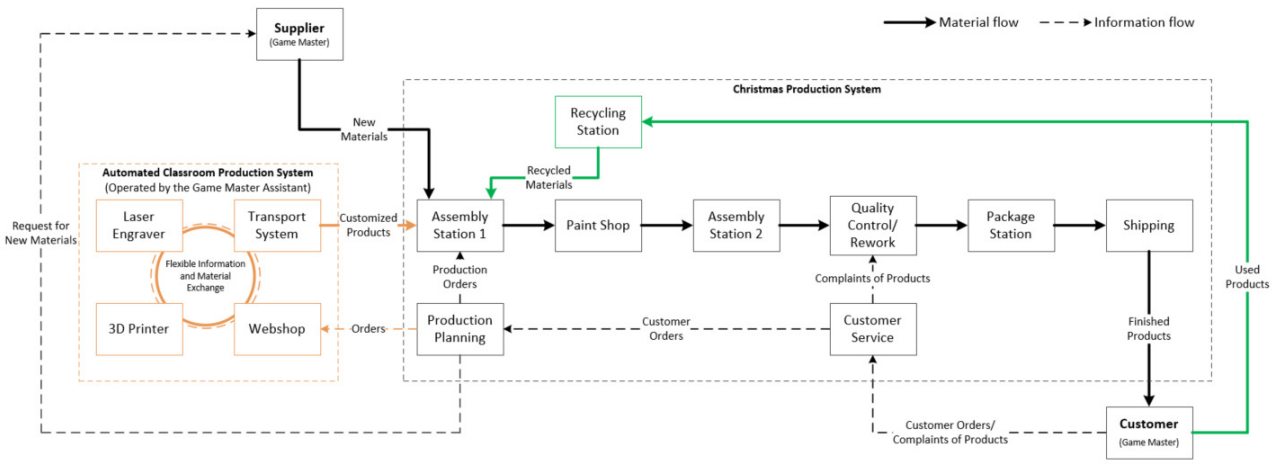

Fig. 4. . Example of one possible game solution of the PSS business game.

The startup pay uses services provided by the "Automated Classroom" and pays for the results. For the use-oriented PSS (green mark), an additional recycling station is required. If the use phase of a Christmas light chain is over, the gamemaster gives the light chains back to the Christmas production system. The recycling station dismantles the light chain and removes faulty and customized materials. The recycled materials are provided to the assembly station to produce new, customer-ready light chains.

These changes in the business game pose several challenges for the students. Due to the new sources for materials, information management is a challenging task. The production scheduler needs an overview of which materials and customized products are available and when to decrease the lead time and safe storage costs. Moreover, the material return and storage is a challenging task. The students need to implement new management processes in the ERP system to solve the challenge.

5. Conclusion and Outlook

This paper shows ways to teach the basics about sustainable PSS through a business game. The target groups for the games are students with production management-oriented fields of studies, e.g., mechanical or industrial engineering students. For this, an analysis was done on the definition and types of PSS and business games in general. An example of how a conventional business game, the in-house-developed "Christmas Production Game," can be adapted according to theories for sustainable PSS is then presented. The enhanced concept of the business game targets the use-oriented and result-oriented PSS. With the aid of the provided tools and information, the students prepare the production for a PSS-applicable environment. With an agile game environment, a high degree of variance is possible. The next steps will be to test the presented concept with students.

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